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Phase III KEYNOTE-789 Study of Pemetrexed and Platinum With or Without Pembrolizumab for Tyrosine Kinase Inhibitor–Resistant, *EGFR*–Mutant, Metastatic Nonsquamous Non–Small Cell Lung Cancer

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ABSTRACT

PURPOSE Epidermal growth factor receptor (*EGFR*) tyrosine kinase inhibitors (TKIs) are standard first-line therapy for *EGFR*-mutant, metastatic non–small cell lung cancer (NSCLC); however, most patients experience disease progression. We report results from the randomized, double-blind, phase III KEYNOTE-789 study of pemetrexed and platinum–based chemotherapy with or without pembrolizumab for TKI-resistant, *EGFR*-mutant, metastatic nonsquamous NSCLC (ClinicalTrials.gov identifier: NCT03515837).

METHODS Adults with pathologically confirmed stage IV nonsquamous NSCLC, documented *DEL19* or *L858R EGFR* mutation, and progression after EGFR-TKI treatment were randomly assigned 1:1 to 35 cycles of pembrolizumab 200 mg or placebo once every 3 weeks plus four cycles of pemetrexed and carboplatin or cisplatin once every 3 weeks and then maintenance pemetrexed. Dual primary end points were progression-free survival (PFS) and overall survival (OS). Final PFS testing was completed at the second interim analysis (IA2; data cutoff, December 3, 2021); OS was tested at final analysis (FA; data cutoff, January 17, 2023). Efficacy boundaries were one-sided $P = .0117$ for PFS and OS.

RESULTS Four hundred ninety-two patients were randomly assigned to pembrolizumab plus chemotherapy (n = 245) or placebo plus chemotherapy (n = 247). At IA2, the median PFS was 5.6 months for pembrolizumab plus chemotherapy versus 5.5 months for placebo plus chemotherapy (hazard ratio [HR], 0.80 [95% CI, 0.65 to 0.97]; $P = .0122$). At FA, the median OS was 15.9 versus 14.7 months, respectively (HR, 0.84 [95% CI, 0.69 to 1.02]; $P = .0362$). Grade ≥3 treatment-related adverse events occurred in 43.7% of pembrolizumab plus chemotherapy recipients versus 38.6% of placebo plus chemotherapy recipients.

CONCLUSION Addition of pembrolizumab to chemotherapy in patients with TKI-resistant, *EGFR*-mutant, metastatic nonsquamous NSCLC did not significantly prolong PFS or OS versus placebo plus chemotherapy in KEYNOTE-789.

ACCOMPANYING CONTENT

Editorial, p. 4003

Appendix

Protocol

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INTRODUCTION

Immune checkpoint inhibitors (ICIs), particularly anti–PD-1 and anti–PD-L1 antibodies, have greatly improved outcomes in advanced or metastatic non–small cell lung cancer (NSCLC).[1] However, patients with mutations in epidermal growth factor receptor (*EGFR*) have been excluded from many of the anti–PD-(L)1-based clinical trials because of initial data suggesting less efficacy of anti–PD-(L)1 monotherapy in this population and the availability of tyrosine kinase inhibitors (TKIs).[2-7] Standard first-line treatment for *EGFR*-mutant NSCLC is third-generation TKIs[2,7]; however,

CONTEXT

Key Objective

The phase III KEYNOTE-789 study examined whether patients with metastatic nonsquamous non–small cell lung cancer (NSCLC) with an epidermal growth factor receptor (EGFR) mutation previously treated with tyrosine kinase inhibitor (TKI) would benefit from treatment with pembrolizumab plus pemetrexed and platinum chemotherapy versus chemotherapy alone.

Knowledge Generated

Pembrolizumab plus pemetrexed-platinum did not significantly prolong progression-free survival or overall survival versus pemetrexed-platinum alone. There were no unexpected toxicities or detrimental impact on quality of life.

Relevance *(T.E. Stinchcombe)*

For patients with EGFR mutant NSCLC with progressive disease after an EGFR TKI, the combination of platinum-pemetrexed and pembrolizumab should not be used as a subsequent therapy.*

*Relevance section written by *JCO* Associate Editor Thomas E. Stinchcombe, MD.

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most patients develop treatment resistance and experience disease progression (PD).[8,9] Subsequent treatment options have limited clinical benefit.[4,6,10,11] These options include local therapy, osimertinib for *T790M*-positive tumors previously untreated with an approved third-generation TKI or previously treated with first- or second-generation TKIs, continuation of the same TKI therapy beyond progression, platinum-based chemotherapy, and atezolizumab (anti–PD-L1) plus bevacizumab (anti–vascular endothelial growth factor [VEGF] antibody) and chemotherapy.[2,7] In addition, single-agent ICI may be considered on failure of TKIs and chemotherapy.[2,7]

Pembrolizumab, an anti–PD-1 monoclonal antibody, as monotherapy[12,13] and in combination with chemotherapy,[14,15] has significantly improved survival outcomes in previously untreated, advanced, or metastatic NSCLC without *EGFR* (or *ALK*) alterations versus chemotherapy alone. Specifically, in the phase III KEYNOTE-189 study, pembrolizumab plus chemotherapy significantly improved overall survival (OS; hazard ratio [HR], 0.49; 95% CI, 0.38 to 0.64; $P < .001$) and progression-free survival (PFS; HR, 0.52 [95% CI, 0.43 to 0.64]; $P < .001$) in previously untreated metastatic nonsquamous NSCLC, irrespective of PD-L1 tumor proportion score (TPS).[14] In light of these data, there was an interest in assessing pembrolizumab plus chemotherapy further in *EGFR*-mutant NSCLC since patients with these mutations were not eligible for inclusion in KEYNOTE-189.

In the phase III KEYNOTE-789 study, we assessed the efficacy and safety of pemetrexed plus platinum chemotherapy with or without pembrolizumab in TKI-resistant, *EGFR*-mutant, metastatic nonsquamous NSCLC, irrespective of PD-L1 TPS (ClinicalTrials.gov: NCT03515837).

METHODS

Study Design and Patients

This randomized, double-blind, phase III study was conducted globally across 139 sites in accordance with Good Clinical Practice guidelines. The study protocol (online only) and amendments were approved by the appropriate ethics body at each site. All patients provided written informed consent before participation. An external data monitoring committee assessed the overall risk-benefit ratio of the trial, safety at periodic intervals, and efficacy at one of the prespecified interim analyses (IAs).

Key eligibility criteria included patients 18 years and older with histologically or cytologically confirmed stage IV nonsquamous NSCLC (per American Joint Committee on Cancer, 8th edition), an Eastern Cooperative Oncology Group performance status of 0 or 1, measurable disease per RECIST version 1.1, tumor tissue sample for assessment, adequate organ function, locally assessed and documented *DEL19* or *L858R EGFR* mutation, and investigator-determined radiographic PD per RECIST version 1.1 after EGFR-TKI treatment. Patients whose disease progressed on first-line osimertinib were eligible regardless of *T790M* mutation status, whereas patients whose disease progressed on erlotinib, afatinib, or gefitinib must have also progressed on osimertinib if *T790M* mutation was detected. Patients with previously treated, radiologically stable brain metastases were eligible.

Treatment

Patients were randomly assigned 1:1 to pembrolizumab 200 mg or saline placebo once every 3 weeks for up to 35 cycles (approximately 2 years). All patients also received pemetrexed

500 mg/m² and investigator's choice of cisplatin 75 mg/m² or carboplatin area under the curve 5 mg/mL/min once every 3 weeks for four cycles followed by pemetrexed 500 mg/m² maintenance therapy. Treatment continued until the maximum number of cycles (except pemetrexed as maintenance) or until PD as confirmed by blinded independent central review (BICR) per RECIST version 1.1, unacceptable toxicity, pregnancy, noncompliance, investigator's decision, or patient withdrawal. Patients assigned to placebo–chemotherapy who experienced PD could cross over to receive pembrolizumab monotherapy on study for a total of 35 cycles if other eligibility criteria were met. Random allocation was performed via an integrated interactive voice- and web-response system and stratified by PD-L1 TPS (<50% *v* ≥ 50%), previous osimertinib (yes *v* no), and region (East Asia *v* non–East Asia).

Assessments

PD-L1 expression was centrally evaluated in archival/newly obtained formalin-fixed, paraffin-embedded tumor samples collected from nonirradiated lesions using PD-L1 IHC 22C3 pharmDx (Agilent Technologies, Carpinteria, CA), and categorized by TPS (percentage of tumor cells with membranous PD-L1 staining). Tumor imaging was performed at baseline, 6 weeks after random assignment, every 9 weeks through 51 weeks, and every 12 weeks thereafter (Appendix 1, online only). In patients who received crossover treatment, tumor imaging was performed within 4 weeks before receiving pembrolizumab and every 12 weeks thereafter. Tumor response was assessed per RECIST version 1.1 by BICR for determination of study end points and eligibility and according to iRECIST[16] by investigator for treatment decisions after PD. Adverse events (AEs) were recorded from random assignment through 30 days (90 days for serious AEs) after last dose of study treatment. AEs were assessed and graded by the investigator according to the National Cancer Institute Common Terminology Criteria for Adverse Events, version 4.0. Survival was assessed every 12 weeks during follow-up. Patient-reported outcomes (PROs) were evaluated using the European Organisation for Research and Treatment of Cancer (EORTC) Quality-of-Life Questionnaire–Core 30 (QLQ-C30), the 13-item EORTC QLQ–Lung Cancer Module (LC13), and the EuroQoL 5-dimension, 5-level questionnaire.

End Points

The dual primary end points were PFS (time from random assignment to first of PD or death) and OS (time from random assignment to death from any cause). Secondary end points included objective response rate (ORR; proportion of patients with complete response [CR] or partial response [PR]), duration of response (DOR; time from evidence of response to first of PD or death in patients with CR or PR), PROs, and safety. The protocol-specified secondary PRO end points included mean score change from baseline to week 18 in the global health status (GHS)/quality of life (QoL) scale (GHS item 29; QoL item 30) and the time to true deterioration (TTD; time from baseline to first onset of ≥10 points of deterioration from baseline, with confirmed second adjacent deterioration of ≥10 points from baseline in any of the three symptoms) in the composite end point of cough (QLQ-LC13 item 1), chest pain (QLQ-LC13 item 10), or dyspnea (QLQ-C30 item 8).

Statistical Analysis

Efficacy was assessed in the intention-to-treat population (all randomly assigned patients), safety in all randomly

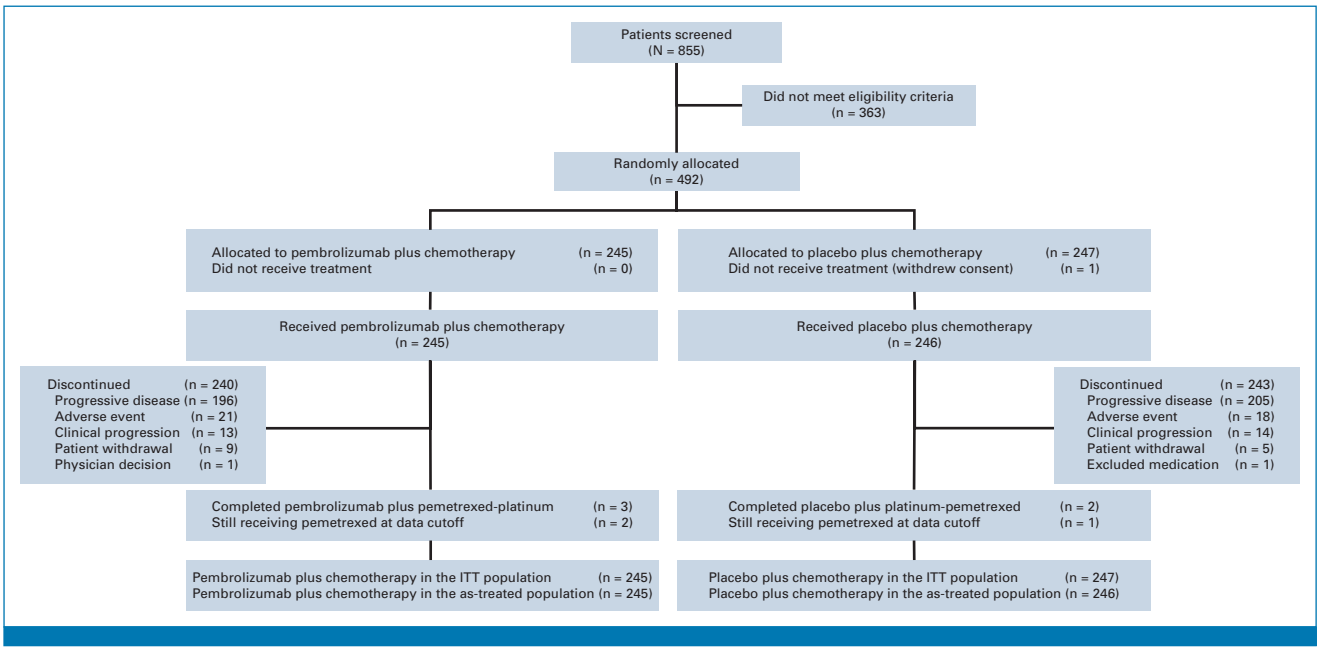

FIG 1. Patient disposition. ITT, intention to treat.

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assigned patients who received ≥1 dose of study treatment, and PROs in all patients with ≥1 PRO assessment available and ≥1 dose of study treatment. PFS, OS, DOR, and TTD were assessed using the Kaplan-Meier method; censoring rules are summarized in the Appendix (Table A1). Between-group differences in OS, PFS, and TTD were assessed using the stratified log-rank test. All *P* values are one-sided. A stratified Cox proportional hazards model with Efron's method of tie handling was used to estimate the magnitude of treatment difference (HR) between treatment groups. Treatment difference in ORR was assessed using a stratified Miettinen and Nurminen method. Change in least squares mean in the GHS/QoL score from baseline to week 18 was assessed using a constrained longitudinal data analysis model, with GHS/QOL score as the response variable and treatment by study visit interaction and stratification factors as covariates. All stratified analyses were performed using the random assignment stratification factors.

The protocol specified three IAs and a final analysis (FA). Second interim analysis (IA2) was the prespecified final PFS analysis. IA3 became the FA because the observed number of OS events at IA3 was close to the target number for FA. The graphical method of Maurer and Bretz was used to control the family-wise type I error rate at 0.025 (one-sided) for hypothesis testing of PFS, OS, and ORR, with preallocated α at .0125, .0125, and 0, respectively (Appendix Fig A1). With 492 patients with ≥24 months of enrollment, the study had approximately 91% power to detect an HR of 0.7 for PFS and approximately 86% power to detect an HR of 0.72 for OS at FA, assuming 423 deaths and 3 IAs, with overall α = .0125 (1-sided). IA1 was planned for approximately 6 months after the last patient was randomly assigned and on the basis of a data cutoff of February 3, 2021. IA2 was planned for approximately 16 months after the last patient was randomly assigned and approximately 414 PFS events were observed. IA2 was based on a data cutoff of December 3, 2021. IA3 (became FA) was planned approximately 29 months after the last patient was randomly assigned and on the basis of a data cutoff of January 17, 2023. Only OS was tested formally at FA. ORR was only to be tested at IA when either PFS or OS was significant. All superiority boundaries were calculated using the Lan-DeMets O'Brien-Fleming spending function. On the basis of the actual number of events observed, the efficacy boundaries were *P* = .0117 for PFS (at IA2) and OS (at FA) per the protocol-specified minimum spending approach.

RESULTS

Patients

Between July 23, 2018, and August 30, 2020, 492 patients were randomly allocated to receive pembrolizumab–chemotherapy (n = 245) or placebo–chemotherapy (n = 247; Fig 1). Demographics and baseline disease characteristics were generally balanced between groups (Table 1). The median age was 63 years, 66.1% of patients were never

TABLE 1. Demographics and Baseline Disease Characteristics

Characteristic	Pembrolizumab Plus Pemetrexed-Platinum (n = 245)	Placebo Plus Pemetrexed-Platinum (n = 247)
Age, years, median (range)	62.0 (34–87)	64.0 (34–84)
Sex		
Male	93 (38.0)	96 (38.9)
Female	152 (62.0)	151 (61.1)
ECOG performance status		
0	71 (29.0)	92 (37.2)
1	174 (71.0)	155 (62.8)
Chemotherapy received		
Cisplatin/pemetrexed	37 (15.1)	43 (17.4)
Carboplatin/pemetrexed	208 (84.9)	203 (82.2)
Missing	0	1 (0.4)
Region		
East Asia	150 (61.2)	150 (60.7)
Non-East Asia	95 (38.8)	97 (39.3)
Smoking status		
Former or current	84 (34.3)	83 (33.6)
Never	161 (65.7)	164 (66.4)
Histology		
Adenocarcinoma	239 (97.6)	243 (98.4)
NSCLC not otherwise specified	5 (2.0)	2 (0.8)
Other	1 (0.4)	2 (0.8)
Brain metastases at baseline	51 (20.8)	47 (19.0)
History of brain metastases	69 (28.2)	69 (27.9)
Liver metastases at baseline	54 (22.0)	40 (16.2)
PD-L1 status		
TPS ≥50%	52 (21.2)	51 (20.6)
TPS 1%-49%	54 (22.0)	72 (29.1)
TPS <1%	127 (51.8)	113 (45.7)
Not evaluable	12 (4.9)	11 (4.5)
Previous use of TKI		
Any first- or second-generation TKI	128 (52.2)	126 (51.0)
First-line osimertinib	28 (11.4)	33 (13.4)
Second-line osimertinib	88 (35.9)	88 (35.6)
Other	1 (0.4)	0
Previous therapy		
Neoadjuvant therapy	1 (0.4)	4 (1.6)
Adjuvant therapy	13 (5.3)	12 (4.9)
Radiation	81 (33.1)	92 (37.2)
T790M mutation		
Positive	95 (38.8)	87 (35.2)
Negative	129 (52.7)	140 (56.7)
Not done	21 (8.6)	20 (8.1)
EGFR-activating mutation		
L858R	103 (42.0)	102 (41.3)
DEL19	139 (56.7)	142 (57.5)
L858R and *DEL19*	2 (0.8)	2 (0.8)
Other	1 (0.4)	1 (0.4)

NOTE. Data are No. (%) unless noted otherwise.
Abbreviations: ECOG, Eastern Cooperative Oncology Group; *EGFR*, epidermal growth factor receptor; NSCLC, non–small cell lung cancer; TKI, tyrosine kinase inhibitor; TPS, tumor proportion score.

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smokers, 48.2% received either first-line or second-line osimertinib, and 20.9% had tumors with PD-L1 TPS ≥50%.

At least one dose of study treatment was received by 245 patients in the pembrolizumab–chemotherapy group and 246 patients in the placebo–chemotherapy group. In the pembrolizumab–chemotherapy group, 240 patients (98.0%) discontinued all treatment, three (1.2%) completed treatment, and two (0.8%) remained on treatment (Fig 1). Eighteen patients (7.3%) began subsequent anti–PD-(L)1 therapy, and 96 (39.2%) were retreated with TKIs. In the placebo–chemotherapy group, 243 patients (98.8%) had discontinued, two (0.8%) completed, and one (0.4%) remained on treatment (Fig 1). At data cutoff, 50 patients (20.2%) from the placebo–chemotherapy group had crossed over to receive pembrolizumab monotherapy on study (Appendix Table A2) and an additional 38 (15.4%) received anti–PD-(L)1 as subsequent therapy outside the study, for an effective crossover rate of 35.6%. In addition, 101 patients (40.9%) in the placebo–chemotherapy group received subsequent TKIs.

Efficacy

At IA2, with a median time from random assignment to data cutoff of 28.6 (16.0 to 40.4) months, 412 events of PD or death

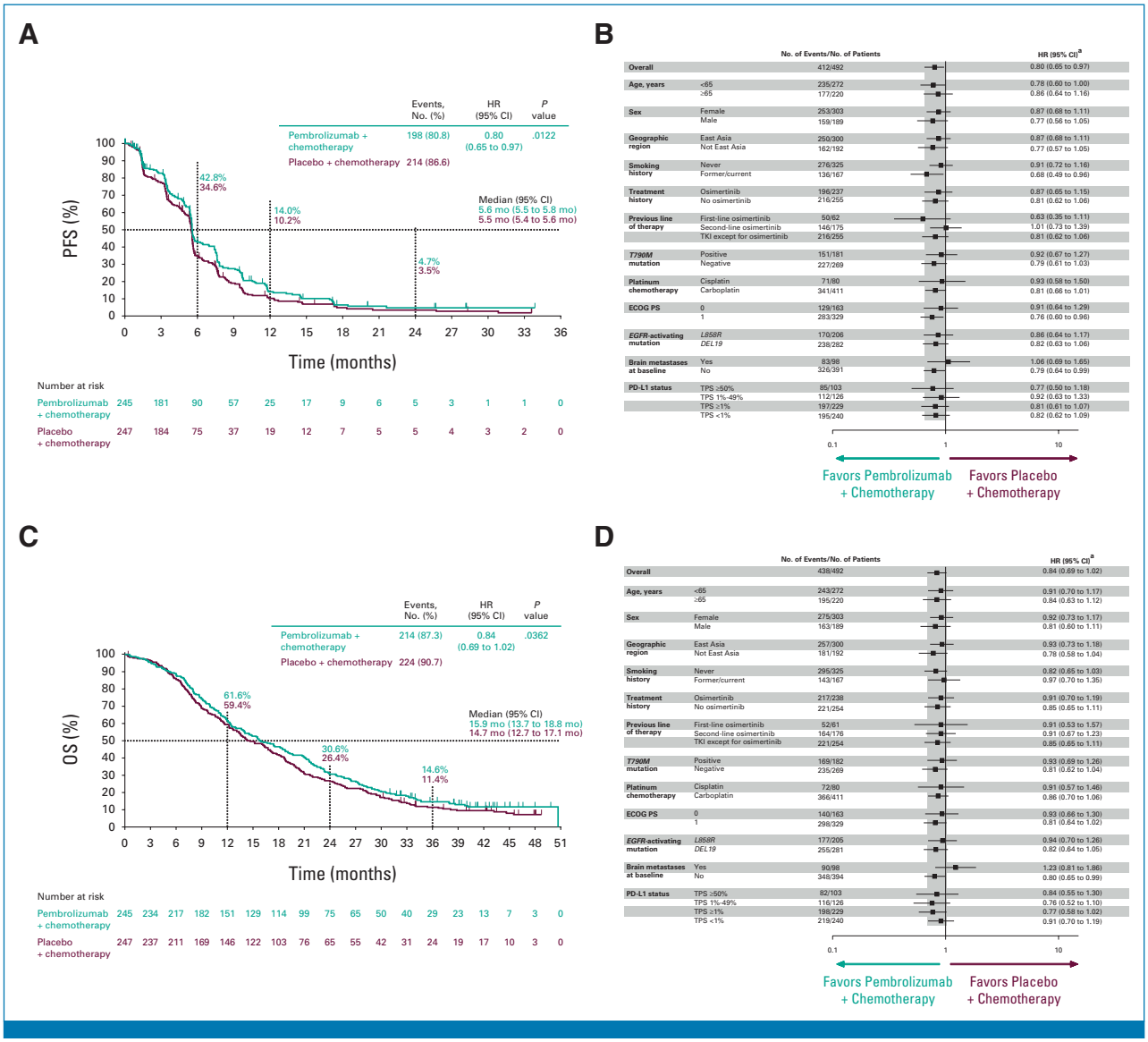

FIG 2. PFS and OS. (A) Kaplan-Meier estimates of PFS tested at IA2 according to RECIST version 1.1 by blinded independent central review. (B) Analysis of PFS in key subgroups. (C) Kaplan-Meier estimates of OS tested at protocol-specified final analysis. (D) Analysis of OS in key subgroups. [a]For the overall population, analysis was based on the Cox regression model with treatment as a covariate stratified by PD-L1 TPS (≥50% or <50%), treatment history (osimertinib or no osimertinib), and geographic region (East Asia or not East Asia). For subgroups, analysis was based on the unstratified Cox regression model with treatment as a covariate. ECOG PS, Eastern Cooperative Oncology Group performance status; *EGFR*, epidermal growth factor receptor; HR, hazard ratio; IA2, second interim analysis; OS, overall survival; PFS, progression-free survival; TKI, tyrosine kinase inhibitor; TPS, tumor proportion score.

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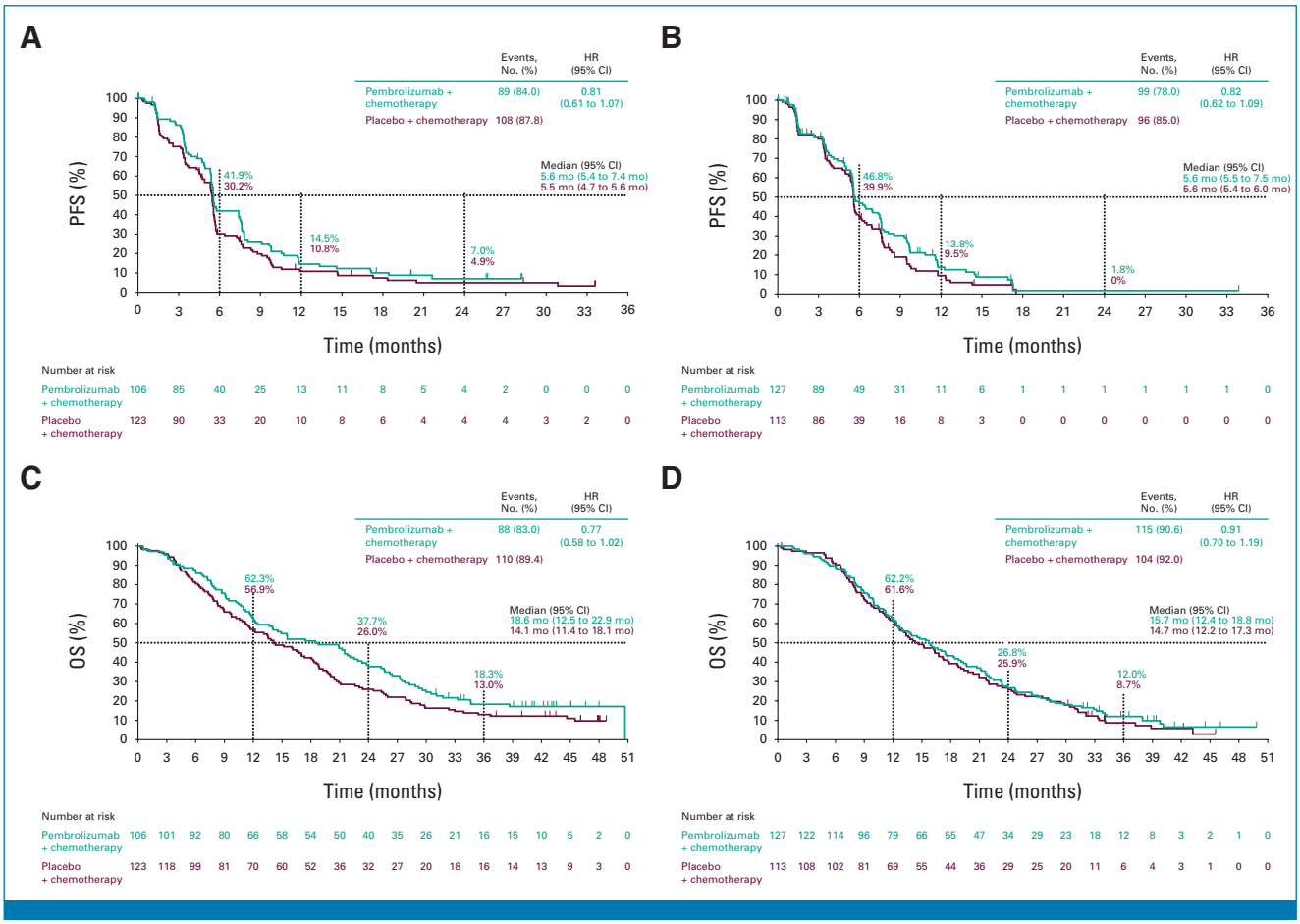

FIG 3. Kaplan-Meier analysis of PFS at IA2 and OS at protocol-specified FA by PD-L1 TPS. (A) PFS in patients with PD-L1 TPS ≥1%. (B) PFS in patients with PD-L1 TPS <1%. (C) OS in patients with PD-L1 TPS ≥1%. (D) OS in patients with PD-L1 TPS <1%. FA, final analysis; HR, hazard ratio; IA2, second interim analysis; OS, overall survival; PFS, progression-free survival; TPS, tumor proportion score.

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had occurred. The median (95% CI) PFS was 5.6 (5.5 to 5.8) months in the pembrolizumab–chemotherapy group versus 5.5 (5.4 to 5.6) months with placebo–chemotherapy. The HR for PFS was 0.80 (95% CI, 0.65 to 0.97; $P = .0122$; Fig 2A). PFS outcomes were similar in most subgroups, including by PD-L1 TPS (Figs 2B, 3A, and 3B, Appendix Figs A2A and A2B).

At FA, the median (range) time from random assignment to data cutoff was 42.0 (29.5 to 53.9) months and there were 438 deaths. The median OS (95% CI) was 15.9 (13.7 to 18.8) months for pembrolizumab–chemotherapy and 14.7 (12.7 to 17.1) months for placebo–chemotherapy. The HR for OS was 0.84 (95% CI, 0.69 to 1.02; $P = .0362$; Fig 2C). OS outcomes were similar across most subgroups analyzed (Fig 2D, Appendix Figs A2C and A2D). There was a trend toward improved OS in patients with PD-L1 TPS ≥1% for pembrolizumab–chemotherapy versus placebo–chemotherapy compared with PD-L1 TPS <1% (Figs 3C and 3D).

At FA, the ORR (95% CI) was 29.0% (23.4 to 35.1) for pembrolizumab–chemotherapy and 27.1% (21.7 to 33.1) for placebo–chemotherapy (Table 2).

In patients who crossed over from placebo–chemotherapy to pembrolizumab monotherapy on study (n = 50), the median PFS (95% CI) from the time of beginning pembrolizumab monotherapy to first documented PD or death was 7.3 (4.8 to 8.5) months.

Safety

The median (range) number of treatment cycles was 8 (1–62) for pembrolizumab–chemotherapy and 8 (1–63) for placebo–chemotherapy. The median duration of treatment was 5.3 (0.03–43.11) months and 5.0 (0.03–42.88) months, respectively. AEs of any grade occurred in 239 of 245 treated patients (97.6%) in the pembrolizumab–chemotherapy group and 241 of 246 patients (98.0%) in the placebo–chemotherapy group (Table 3). Treatment-related AEs occurred in 220 patients (89.8%) and 212 (86.2%), respectively. These were grade ≥3 in 107 (43.7%) and 95 (38.6%) patients, respectively. There was one fatal treatment-related AE due to myocarditis in the pembrolizumab–chemotherapy group and two in the placebo–chemotherapy group due to bone marrow failure and general physical health deterioration.

TABLE 2. Summary of Confirmed Tumor Response Assessed per RECIST Version 1.1 by Blinded Independent Central Review (ITT Population)

Response	Pembrolizumab Plus Pemetrexed-Platinum (n = 245)	Placebo Plus Pemetrexed-Platinum (n = 247)
ORR (95% CI), %	29.0 (23.4 to 35.1)	27.1 (21.7 to 33.1)
Best overall response, No. (%)		
CR	5 (2.0)	3 (1.2)
PR	66 (26.9)	64 (25.9)
SD[a]	121 (49.4)	117 (47.4)
PD	37 (15.1)	52 (21.1)
Not evaluable[b]	8 (3.3)	5 (2.0)
No assessment[c]	8 (3.3)	6 (2.4)
DOR,[d] median (range), months	6.3 (2.3-40.8+)[e]	5.6 (1.8+-40.6+)[e]
DOR ≥9 months,[d] No. (%)	23 (34.0)	13 (22.9)
Time to response, median (range), months	1.5 (1.2-9.7)	1.5 (1.2-25.3)

Abbreviations: CR, complete response; DOR, duration of response; ITT, intention to treat; ORR, objective response rate; PD, progressive disease; PR, partial response; SD, stable disease.
[a]Stable disease includes stable disease, non-CR/non-PD, and no evidence of disease (no lesions identified at baseline or postbaseline assessments).
[b]Postbaseline assessment(s) available but not evaluable or CR/PR/SD <6 weeks from random assignment.
[c]No postbaseline assessment available for response evaluation.
[d]On the basis of Kaplan-Meier estimate.
[e]The + indicates no progressive disease by the time of last disease assessment.

Immune-mediated AEs and infusion reactions occurred in 49 patients (20.0%) who received pembrolizumab–chemotherapy and 20 (8.1%) in the placebo–chemotherapy group. These were grade ≥3 in 11 (4.5%) and five (2.0%) patients, respectively. The only fatal immune-mediated event was myocarditis (0.4%; noted above) in the pembrolizumab–chemotherapy group.

PROs

The PRO analysis population comprised 473 patients (pembrolizumab–chemotherapy, n = 237; placebo–chemotherapy, n = 236) who completed ≥1 QLQ-C30 and QLQ-LC13 assessment. At baseline, the completion and compliance rates were >97.0% for each treatment group and >71.2% at week 18 (Appendix Table A3).

At week 18, the least squares mean (95% CI) change from baseline in QLQ-C30 GHS/QoL scores was stable in both treatment groups (Appendix Table A4); no clinically meaningful between-group differences were observed (least squares mean difference 1.59 points; 95% CI, −1.93 to 5.10). The median (95% CI) TTD for the composite end point was not reached (10.05 months to not reached) for pembrolizumab–chemotherapy and 17.97 months (6.67 months to not reached) for placebo–chemotherapy (HR, 0.93 [95% CI, 0.68 to 1.27]; Fig 4).

DISCUSSION

In this study of TKI-resistant, *EGFR*-mutant, metastatic nonsquamous NSCLC, the addition of pembrolizumab to pemetrexed and platinum chemotherapy did not significantly prolong PFS and OS versus placebo plus chemotherapy per the prespecified statistical analysis plan. No new safety signals were identified in either treatment group. Pembrolizumab–chemotherapy demonstrated no clinically meaningful detriment in overall QoL and comparable TTD in disease-related symptoms versus placebo–chemotherapy. Among 50 patients who received pembrolizumab in the on-study crossover, 40% had PD-L1 TPS <1%. There was evidence of antitumor activity in these patients; however, no meaningful conclusions can be drawn because of small sample size and the non-prespecified nature of this analysis.

PFS and OS outcomes were similar in most subgroups assessed. No specific differences were observed in outcomes by choice of platinum chemotherapy (carboplatin or cisplatin), similar to other studies examining immunotherapy plus chemotherapy.[14,17] There was a trend toward longer PFS and OS with pembrolizumab–chemotherapy versus placebo–chemotherapy in patients without brain metastasis at baseline; albeit, a similar trend was not observed in patients with brain metastasis at baseline, which represented a small subgroup with wide CIs. Trends in PD-L1 TPS subgroups also differed, with outcomes in patients with PD-L1 TPS ≥50% being similar between treatment groups, but a trend for improved OS with pembrolizumab–chemotherapy in patients with PD-L1 TPS ≥1% versus <1%. However, a similar trend was not observed for PFS. Importantly, these subgroup analyses were not adjusted for multiplicity, and no definitive conclusions can be drawn from these observations. Similar to these findings, previous studies of ICIs in *EGFR*-mutant NSCLC have shown a variable relationship between PD-L1 expression and outcomes to therapy, suggesting that PD-L1 remains an inconclusive biomarker for response in this setting.[6,18-21] Overall, *EGFR*-mutant tumors have been documented to have immunosuppressive or cold microenvironments and lower tumor mutational burden compared with tumors without *EGFR* mutations and correlate with inferior response to anti–PD-(L)1 therapy.[11,18]

Results presented here follow those from phase II studies of pembrolizumab plus chemotherapy in TKI-resistant advanced *EGFR*-mutant NSCLC where the ORR ranged from 42% to 58%.[22,23] However, these benefits were not realized in our larger and more robust study. Similarly, in CheckMate 722, a prospective phase III study of TKI-resistant, *EGFR*-mutant, metastatic NSCLC, there was no significant improvement in PFS (primary end point) with nivolumab (anti–PD-1) plus chemotherapy versus chemotherapy alone (HR, 0.75 [95% CI, 0.56 to 1.00]).[24] However, the study did

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TABLE 3. AEs in Patients Who Received ≥1 Dose of Study Treatment

Event	Pembrolizumab Plus Pemetrexed-Platinum (n = 245), No. (%)		Placebo Plus Pemetrexed-Platinum (n = 246), No. (%)	
Any AE (all-cause)	239 (97.6)		241 (98.0)	
Grade 3 to 5	137 (55.9)		143 (58.1)	
Led to death[a]	5 (2.0)		12 (4.9)	
Treatment-related	220 (89.8)		212 (86.2)	
Grade 3 to 5	107 (43.7)		95 (38.6)	
Led to discontinuation of any treatment component	40 (16.3)		29 (11.8)	
Led to discontinuation of pembrolizumab or placebo	24 (9.8)		11 (4.5)	
Led to discontinuation of any chemotherapy	31 (12.7)		29 (11.8)	
Led to discontinuation of all treatment components	7 (2.9)		5 (2.0)	
Treatment-related AEs occurring in ≥10% of patients in either treatment group	Any Grade	Grade 3–5	Any Grade	Grade 3–5
Anemia	94 (38.4)	31 (12.7)	99 (40.2)	23 (9.3)
Decreased neutrophil count	92 (37.6)	56 (22.9)	111 (45.1)	52 (21.1)
Nausea	85 (34.7)	5 (2.0)	102 (41.5)	8 (3.3)
Decreased white blood cell count	74 (30.2)	26 (10.6)	86 (35.0)	22 (8.9)
Fatigue	57 (23.3)	8 (3.3)	44 (17.9)	5 (2.0)
Increased alanine aminotransferase	55 (22.4)	10 (4.1)	51 (20.7)	6 (2.4)
Increased aspartate aminotransferase	54 (22.0)	4 (1.6)	50 (20.3)	2 (0.8)
Decreased platelet count	54 (22.0)	21 (8.6)	49 (19.9)	13 (5.3)
Decreased appetite	53 (21.6)	7 (2.9)	49 (19.9)	1 (0.4)
Constipation	47 (19.2)	2 (0.8)	41 (16.7)	1 (0.4)
Vomiting	36 (14.7)	2 (0.8)	34 (13.8)	6 (2.4)
Asthenia	25 (10.2)	5 (2.0)	20 (8.1)	1 (0.4)
Immune-mediated AEs and infusion reactions	49 (20.0)	11 (4.5)	20 (8.1)	5 (2.0)
Hypothyroidism	14 (5.7)	0	6 (2.4)	0
Hyperthyroidism	13 (5.3)	0	3 (1.2)	0
Infusion reactions	5 (2.0)	0	2 (0.8)	0
Pneumonitis	5 (2.0)	2 (0.8)	4 (1.6)	2 (0.8)
Adrenal insufficiency	2 (0.8)	0	0	0
Hepatitis	2 (0.8)	2 (0.8)	1 (0.4)	1 (0.4)
Myositis	2 (0.8)	2 (0.8)	1 (0.4)	0
Nephritis	2 (0.8)	1 (0.4)	1 (0.4)	1 (0.4)
Severe skin reactions	2 (0.8)	2 (0.8)	0	0
Thyroiditis	2 (0.8)	0	3 (1.2)	0
Colitis	1 (0.4)	1 (0.4)	1 (0.4)	1 (0.4)
Hypophysitis	1 (0.4)	0	0	0
Myelitis	1 (0.4)	0	0	0
Myocarditis	1 (0.4)	1 (0.4)	0	0
Type 1 diabetes mellitus	1 (0.4)	1 (0.4)	0	0

Abbreviation: AE, adverse event.

[a]One patient in the pembrolizumab plus pemetrexed-platinum group died because of a treatment-related AE of myocarditis; two patients in the placebo plus pemetrexed-platinum group died because of treatment-related AEs of bone marrow failure (n = 1) and general physical health deterioration (n = 1).

not reach the required number of PFS events and was therefore underpowered.[24]

By contrast, the phase III ORIENT-31 study showed significantly improved PFS outcomes (primary end point) in TKI-resistant, *EGFR*-mutant advanced or metastatic NSCLC, with sintilimab (anti–PD-1) plus chemotherapy versus chemotherapy alone ($P = .016$), but without similar benefit in OS.[25] Comparatively greater benefit was observed in a third treatment arm that added IBI305 (anti-VEGF) to sintilimab plus chemotherapy versus chemotherapy alone ($P < .0001$) although no direct comparison between all three study arms

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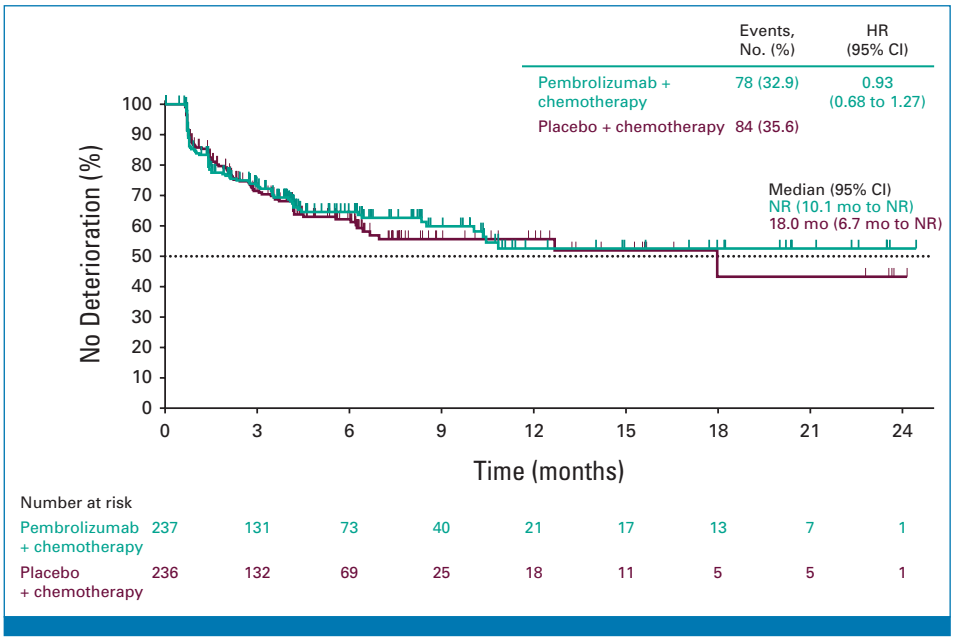

FIG 4. Kaplan-Meier estimate of time to true deterioration for the composite end point of cough, chest pain, and dyspnea at the protocol-specified final analysis. HR, hazard ratio; NR, not reached.

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has been reported.[26] Similarly, the ATTLAS study showed significant PFS benefit (primary end point) with atezolizumab plus bevacizumab and chemotherapy versus chemotherapy alone (P = .004) in patients with *EGFR*- or *ALK*-mutant metastatic NSCLC, but without OS benefit.[21] These results built on a subgroup analysis (n = 124) of *EGFR*-mutant NSCLC from the phase III IMpower150 study of atezolizumab plus bevacizumab and chemotherapy, which showed a trend toward improved OS versus bevacizumab plus chemotherapy alone.[10] Notably, no OS or PFS benefit was observed with atezolizumab plus chemotherapy (without bevacizumab) versus bevacizumab plus chemotherapy.[10] While these studies suggest an added benefit of anti–VEGF therapy when combined with anti–PD-(L)1 therapy and chemotherapy, there were no statistically significant improvements in OS outcomes. Differences in outcomes between KEYNOTE-789 and these studies may be explained by several factors. First, moderate clinical benefit from anti-VEGF therapy has been reported for *EGFR*-mutant NSCLC, albeit with some increased toxicity.[27-29] Other factors include different study demographics (ie, ORIENT-31 was conducted in China, and ATTLAS in South Korea) and previous osimertinib treatment. In CheckMate 722, for example, 23% of patients received previous osimertinib versus 48% in KEYNOTE-789.[24] Similarly, 11% of patients received previous third-generation TKIs (including osimertinib) in ORIENT-31[25] and only 10 patients received previous osimertinib in IMpower150.[10] Because third-generation TKIs are the standard of care for these patients, the results from KEYNOTE-789 may more accurately reflect the real-world situation. In addition, treatment with TKIs can lead to a range of resistance mechanisms, changes in the tumor microenvironment to become more immunosuppressive, and decreased PD-L1 mRNA expression and PD-L1 degradation, which could decrease the response to anti–PD-(L)1-based therapies.[30-32] There were no new safety signals with pembrolizumab–chemotherapy in KEYNOTE-789. The incidence of grade 3 to 5 treatment-related AEs was similar between treatment groups, as was the proportion of patients who discontinued treatment because of AEs. The safety profile of pembrolizumab–chemotherapy in KEYNOTE-789 was also comparable with that observed in patients with previously untreated, metastatic NSCLC without *EGFR* or *ALK* alterations.[14,15]

In conclusion, pembrolizumab plus pemetrexed and platinum chemotherapy did not significantly prolong PFS and OS versus chemotherapy alone in patients with *EGFR*-mutant, TKI-resistant metastatic nonsquamous NSCLC, per the prespecified statistical analysis plan. There were no unexpected toxicities, and no detrimental impact in QoL was associated with pembrolizumab–chemotherapy compared with chemotherapy alone. Evidence to date suggests that anti–PD-(L)1 plus chemotherapy may not provide sufficient clinical benefit compared with chemotherapy alone, and a better understanding of potential biomarkers may be required to identify patients who will experience the full benefit of anti–PD-(L)1 therapy in TKI-resistant, *EGFR*-mutant, metastatic nonsquamous NSCLC.

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DATA SHARING STATEMENT

Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc, Rahway, NJ, USA (MSD), is committed to providing qualified scientific researchers access to anonymized data and clinical study reports from the company's clinical trials for the purpose of conducting legitimate scientific research. MSD is also obligated to protect the rights and privacy of trial participants and, as such, has a procedure in place for evaluating and fulfilling requests for sharing company clinical trial data with qualified external scientific researchers. The MSD data sharing website (available at http://engagezone.msd.com/ds_documentation.php) outlines the process and requirements for submitting a data request. Applications will be promptly assessed for completeness and policy compliance. Feasible requests will be reviewed by a committee of MSD subject matter experts to assess the scientific validity of the request and the qualifications of the requestors. In line with data privacy legislation, submitters of approved requests must enter into a standard data-sharing agreement with MSD before data access is granted. Data will be made available for request after product approval in the United States and EU or after product development is discontinued. There are circumstances that may prevent MSD from sharing requested data, including country- or region-specific regulations. If the request is declined, it will be communicated to the investigator. Access to genetic or exploratory biomarker data requires a detailed, hypothesis-driven statistical analysis plan that is collaboratively developed by the requestor and MSD subject matter experts; after approval of the statistical analysis plan and execution of a data-sharing agreement, MSD will either perform the proposed analyses and share the results with the requestor or construct biomarker covariates and add them to a file with clinical data that is uploaded to an analysis portal so that the requestor can perform the proposed analyses.

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Phase III KEYNOTE-789 Study of Pemetrexed and Platinum With or Without Pembrolizumab for Tyrosine Kinase Inhibitor–Resistant, *EGFR*–Mutant, Metastatic Nonsquamous Non–Small Cell Lung Cancer

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APPENDIX. DESCRIPTION OF BRAIN METASTASIS IMAGING REQUIREMENTS IN KEYNOTE-789

Brain magnetic resonance imaging was required for all patients at screening, unless medically contraindicated; computed tomography with contrast was allowed in those instances. Ideally, for a given patient, the same type of scan, same scanner, and the use of contrast were recommended throughout the study to optimize the reproducibility and accuracy of the assessment of existing and new tumor burden. All images were submitted to the central imaging vendor, and radiographic progression per RECIST version 1.1 was confirmed by the central imaging vendor. Imaging assessments were performed at 6 weeks from the date of random assignment, then every 9 weeks until week 51 (or more frequently if clinically indicated), and every 12 weeks thereafter. For imaging throughout the study, on-study brain or bone imaging was performed if clinically indicated or to confirm complete response (if other lesions indicated complete response and brain or bone lesions existed at baseline).

TABLE A1. Censoring Rules for Primary Kaplan-Meier Analyses

Situation	Censoring Rule
Progression-free survival	
PD or death documented after ≤1 missed disease assessment, and before any new anticancer therapy, if any	Progressed at the date of documented PD or death
PD or death documented immediately after ≥2 consecutive missed disease assessments or after new anticancer therapy, if any	Censored at last disease assessment before the earlier date of ≥2 consecutive missed disease assessments and new anticancer therapy, if any
No PD, no death; new anticancer treatment is not initiated	Censored at last disease assessment
No PD, no death; new anticancer treatment is initiated	Censored at last disease assessment before new anticancer treatment
Overall survival	
No death	Censored at the last date patient was known to be alive
Duration of response[a]	
No progression nor death, no new anticancer treatment initiated	Censored at last adequate disease assessment
No progression nor death, new anticancer treatment initiated	Censored at last adequate disease assessment before new anticancer therapy
Death or progression immediately after ≥2 consecutive missed disease assessments or after new anticancer therapy, if any	Censored at the earlier date of last adequate disease assessment before ≥2 missed adequate disease assessments and new anticancer therapy, if any
Death or progression after ≤1 missed disease assessment and before the new anticancer therapy, if any	End of response at the date of PD or death
Time to deterioration	
Deterioration confirmed	Event observed at the time of assessment (first deterioration)
Ongoing or discontinued from the study without confirmed deterioration	Right censored at the time of last assessment
No baseline assessments	Right censored at treatment start date

Abbreviation: PD, progressive disease.

[a]For duration of response, a missed disease assessment includes any assessment that is not obtained or is considered inadequate for evaluation of response.

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TABLE A2. Demographics and Baseline Disease Characteristics of Patients Who Crossed Over to Pembrolizumab Monotherapy From Placebo Plus Pemetrexed-Platinum

Characteristic	Patients Who Crossed Over to Pembrolizumab From Placebo (n = 50)
Age, years, median (range)	65.0 (45-79)
Sex	
Male	18 (36.0)
Female	32 (64.0)
ECOG performance status	
0	17 (34.0)
1	33 (66.0)
Chemotherapy received	
Cisplatin/pemetrexed	11 (22.0)
Carboplatin/pemetrexed	39 (78.0)
Region	
East Asia	38 (76.0)
Non-East Asia	12 (24.0)
Smoking status	
Former or current	15 (30.0)
Never	35 (70.0)
Adenocarcinoma histology	50 (100.0)
Brain metastases at baseline	9 (18.0)
History of brain metastases	11 (22.0)
Liver metastases at baseline	11 (22.0)
PD-L1 status	
TPS ≥50%	13 (26.0)
TPS 1%-49%	15 (30.0)
TPS <1%	20 (40.0)
Not evaluable	2 (4.0)
Previous use of TKI	
Any first or second generation TKI	28 (56.0)
First-line osimertinib	4 (8.0)
Second-line osimertinib	18 (36.0)
Previous therapy	
Neoadjuvant therapy	1 (2.0)
Adjuvant therapy	2 (4.0)
Radiation	14 (28.0)
T790M mutation	
Positive	18 (36.0)
Negative	29 (58.0)
Not done	3 (6.0)
EGFR-activating mutation	
L858R	17 (34.0)
DEL19	33 (66.0)

NOTE. Data are No. (%) unless noted otherwise.
Abbreviations: ECOG, Eastern Cooperative Oncology Group; *EGFR*, epidermal growth factor receptor; NSCLC, non–small cell lung cancer; TKI, tyrosine kinase inhibitor; TPS, tumor proportion score.

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TABLE A3. Completion and Compliance Rates for EORTC QLQ-C30 and QLQ-LC13

	QLQ-C30		QLQ-LC13	
Treatment Visit	Pembrolizumab Plus Pemetrexed-Platinum (n = 241)	Placebo Plus Pemetrexed-Platinum (n = 243)	Pembrolizumab Plus Pemetrexed-Platinum (n = 241)	Placebo Plus Pemetrexed-Platinum (n = 243)
Baseline				
Completion	237/241 (98.3)	236/243 (97.1)	237/241 (98.3)	236/243 (97.1)
Compliance	237/241 (98.3)	236/243 (97.1)	237/241 (98.3)	236/243 (97.1)
Week 3				
Completion	209/241 (86.7)	219/243 (90.1)	208/241 (86.3)	219/243 (90.1)
Compliance	209/223 (93.7)	219/228 (96.1)	208/223 (93.3)	219/228 (96.1)
Week 6				
Completion	198/241 (82.2)	205/243 (84.4)	196/241 (81.3)	205/243 (84.4)
Compliance	198/216 (91.7)	205/220 (93.2)	196/216 (90.7)	205/220 (93.2)
Week 9				
Completion	181/241 (75.1)	195/243 (80.2)	182/241 (75.5)	194/243 (79.8)
Compliance	181/196 (92.3)	195/213 (91.5)	182/196 (92.9)	194/213 (91.1)
Week 12				
Completion	175/241 (72.6)	177/243 (72.8)	174/241 (72.2)	177/243 (72.8)
Compliance	175/183 (95.6)	177/191 (92.7)	174/183 (95.1)	177/191 (92.7)
Week 15				
Completion	165/241 (68.5)	167/243 (68.7)	164/241 (68.0)	165/243 (67.9)
Compliance	165/177 (93.2)	167/186 (89.8)	164/177 (92.7)	165/186 (88.7)
Week 18				
Completion	173/241 (71.8)	173/243 (71.2)	171/241 (71.0)	173/243 (71.2)
Compliance	173/179 (96.6)	173/178 (97.2)	171/179 (95.5)	173/178 (97.2)
Week 27				
Completion	127/241 (52.7)	121/243 (49.8)	127/241 (52.7)	120/243 (49.4)
Compliance	127/147 (86.4)	121/147 (82.3)	127/147 (86.4)	120/147 (81.6)
Week 36				
Completion	89/241 (36.9)	77/243 (31.7)	88/241 (36.5)	77/243 (31.7)
Compliance	89/100 (89.0)	77/93 (82.8)	88/100 (88.0)	77/93 (82.8)
Week 45				
Completion	61/241 (25.3)	49/243 (20.2)	61/241 (25.3)	49/243 (20.2)
Compliance	61/74 (82.4)	49/55 (89.1)	61/74 (82.4)	49/55 (89.1)
Week 54				
Completion	46/241 (19.1)	34/243 (14.0)	46/241 (19.1)	33/243 (13.6)
Compliance	46/52 (88.5)	34/40 (85.0)	46/52 (88.5)	33/40 (82.5)

NOTE. Completion was defined as the proportion of patients who completed a PRO assessment among the total PRO analysis population. Compliance was defined as the proportion of patients who completed a PRO assessment among those expected to complete the instruments at each visit (excluding patients who discontinued study treatment).

Abbreviations: EORTC, European Organisation for Research and Treatment of Cancer; PRO, patient-reported outcome; QLQ-C30, Quality-of-Life Questionnaire—Core 30; QLQ-LC13, 13-item EORTC QLQ—Lung Cancer Module.

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TABLE A4. Change From Baseline to Week 18 in the EORTC QLQ-C30 GHS/QoL Scale

Time Point	Pembrolizumab Plus Pemetrexed-Platinum (n = 241)	Placebo Plus Pemetrexed-Platinum (n = 243)
Baseline, No.[a]	237	236
Mean (SD) score	65.65 (20.88)	67.30 (19.65)
Week 18, No.[a]	172	173
Mean (SD) score	68.65 (17.44)	66.91 (18.12)
Change from baseline, No.[b]	241	243
LS mean score (95% CI)	−0.46 (−3.17 to 2.25)	−2.05 (−4.74 to 0.64)
Between-group difference in LS mean (95% CI)	1.59 (−1.93 to 5.10)	

Abbreviations: EORTC, European Organisation for Research and Treatment of Cancer; GHS/QoL, global health status/quality of life; QLQ-C30, Quality-of-Life Questionnaire–Core 30.
[a]No. represents the number of patients in each treatment group with nonmissing assessments at the specific time point.
[b]No. represents the number of patients in the analysis population in each treatment group.

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TABLE A5. List of Investigators

Country/Region	Principal Investigator
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	Rina Hui/Mark Wong
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	Sergio Jobim de Azevedo
	Fabio Andre Franke
	Joilda Batista De Almeida Rego
	Pedro Rafael Martins De Marchi/Gustavo Dix Junqueira
	Fernanda Maris Peria
	Leandro Brust
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	Mark Doherty/Ambika Parmar/Ines B. Menjak
	Natasha B. Leighl
	Jason Agulnik
China	Shun Lu
	Zhigang Han
	Jiuwei Cui
	Li Zhang
	Ying Cheng
	Gongyan Chen
	Helong Zhang
	Yu Yao
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	Qiming Wang
	Xin Zhang/Yong Zhang
	Jianying Zhou
	Kejing Ying
	Yun Fan
	Yan Wang
	Ziping Wang
	Jifeng Feng
	YingYing Du
	Lin Wu
	Cheng Huang
	Xiangdong Zhou
France	Maurice Perol
	Julien Domont
	Corinne Lamour/Julien Dutilh
	Youssef Oulkhouir
	Virginie Westeel
	Delphine Carmier
	Bruno Coudert/Aurelie Lagrange
	Dominique Spaeth
	Stanislas Ropert

(continued in next column)

TABLE A5. List of Investigators (continued)

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	Martin Wermke
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	Annette Mueller
	Gunther Vogel
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	Chung Man James Ho
	Siu Hong Oscar Chan
	Sing Hung Lo/Shi Feng Jonathan Nyaw
	Yu Chung Jacky Li
Israel	Jair Bar
	Maya Gottfried
	Julia Dudnik
	Alona Zer/Mor Moskovitz/Mirjana Wollner
	Ofer Rotem
	Sivan Shamai
	Noam Asna
	Mhameed Kamel
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	Francesco Di Costanzo/Laura Doni/Francesca Mazzoni
	Francesco Ferrau
	Filippo de Marinis
	Giuseppe Tonini
	Domenico Galetta
	Francovito Piantedosi/Fabiana Vitiello
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	Toshiaki Takahashi
	Yuichiro Ohe
	Yoshihiro Hattori
	Isamu Okamoto
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	Hideo Saka/Masahide Oki
	Terufumi Kato
	Hiroshi Tanaka
	Toru Kumagai/Takako Inoue
	Toyoaki Hida/Yoshitsugu Horio
	Shunsuke Teraoka
	Eiki Ichihara
	Kazuma Kishi/Hisashi Takaya/Daiya Takai
	Toshiyuki Kozuki
	Kazuo Kasahara/Yuichi Tambo
	Yukio Hosomi
	Masashi Kondo
	Masao Ichiki/Hiroaki Takeoka

(continued on following page)

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TABLE A5. List of Investigators (continued)

Country/Region	Principal Investigator
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	Oscar Gerardo Arrieta Rodriguez
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	Keunchil Park/Se-Hoon Lee
	Ki Hyeong Lee
	Jong-Seok Lee
	Jin Hyoung Kang
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	Eun Kyung Cho
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	Pilar Garrido Lopez
	Margarita Majem Tarruella
	Luis Paz-Ares Rodriguez
	David Vicente Baz
	Enriqueta Felip Font
	Manuel Cobo-Dols
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	Bengt Bergman
	Ronny Ohman
	Anders Vikstrom
Taiwan	Chih-Hsin Yang
	Chao-Hua Chiu/Hsu-Ching Huang
	Cheng-Ta Yang
	Jian Su
	Gee-Chen Chang/Tsung-Ying Yang
	Te-Chun Hsia
	Wu-Chou Su/Shang-Yin Wu
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	Chih-Bin Lin
	Jih-Hsiang Lee
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	Jonathan Shamash
	Dionysis Papadatos-Pastos
	Pooja Jain
	Melanie Jane Mackean

(continued in next column)

TABLE A5. List of Investigators (continued)

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	Smitha Menon
	Ahmad Raza/Li Ge
	Gregory J. Riely
	Nagashree Seetharamu
	Laura Stampleman
	Janakiraman Subramanian
	Donald B. Wender
	Ronald B. Natale
	Viola Zhu/Sai-Hong Ignatius Ou
	Rachel Sanborn
	Makenzi C. Evangelist

NOTE. Where two names are listed, original investigator's name is followed by subsequent investigator's name.

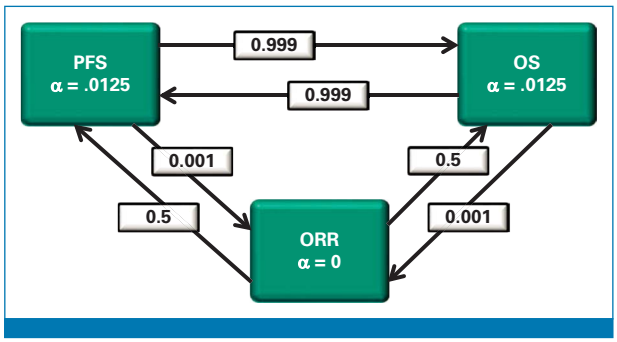

FIG A1. Multiplicity diagram for type I error control. The initial α allocated to each hypothesis is represented in the green boxes with the weights for reallocation from each hypothesis to others represented in the white boxes. If both the PFS and OS hypotheses are rejected, the reallocation strategy allows retesting of ORR at α = .025 on the basis of the *P* value at IA1. The actual boundaries were updated on the basis of the actual number of events observed and calculated using the Lan-DeMets O'Brien-Fleming spending function with the information fraction on the basis of the minimum of the expected information fraction and the actual information fraction at each analysis. ORR, objective response rate; OS, overall survival; PFS, progression-free survival.

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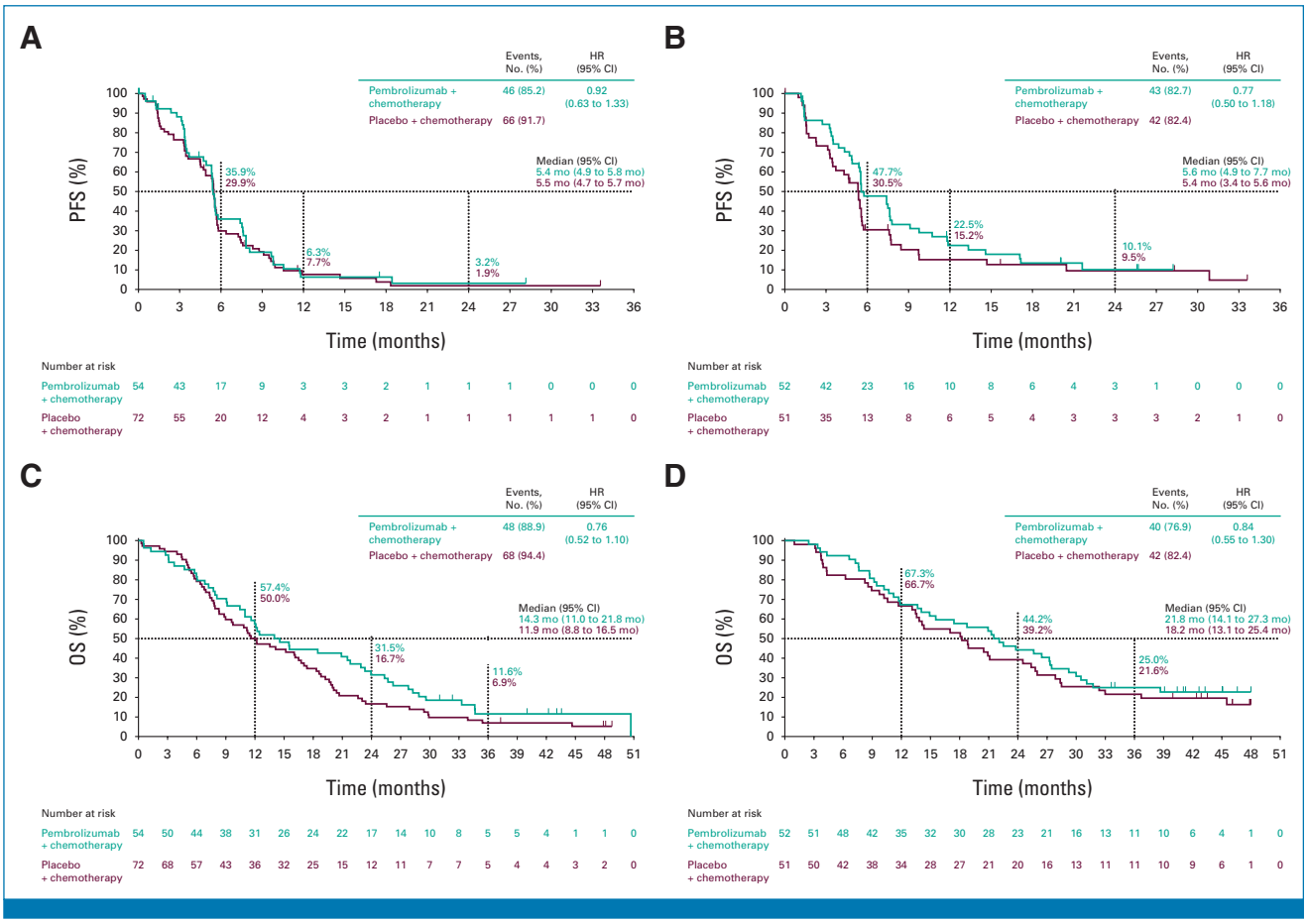

FIG A2. Kaplan-Meier analysis of progression-free survival per RECIST version 1.1 by BICR at IA2 and OS at the final analysis in PD-L1 TPS subgroups. (A) PFS in patients with PD-L1 TPS 1%-49%. (B) PFS in patients with PD-L1 TPS ≥50%. (C) OS in patients with PD-L1 1%-49%. (D) OS in patients with PD-L1 TPS ≥50%. BICR, blinded independent central review; HR, hazard ratio; IA2, second interim analysis; OS, overall survival; PFS, progression-free survival; TPS, tumor proportion score.

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